AI-Based Fertilizer Recommendation System

Based on Soil Test Values using Random Forest Algorithm

Course: Application of AI & ML in Agriculture
University: UAS Bengaluru

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Introduction

Agriculture: Backbone of India's Economy

- Crop productivity depends on proper fertilizer use
- Farmers rely on experience, not data
- Leads to overuse → soil damage
- Leads to underuse → low yield

AI Solution Benefits:

✓ Predict exact nutrient needs

✓ Improve decision-making

✓ Use ML (Random Forest)

✓ Recommend based on soil data

⚠ Problem Statement

Current Issues:

- ✗ Lack of scientific fertilizer recommendations
- ✗ Imbalanced nutrient application (NPK)
- ✗ Soil fertility degradation
- ✗ Increased cost of cultivation
- ✗ Reduced crop yield

Need for Smart System that Uses Soil Data for Accurate Recommendations

Motivation

Solve Real-World Problems

Bridge traditional and modern farming

Help Farmers

Reduce input cost & increase productivity

Promote Sustainability

Precision & eco-friendly agriculture

Boost Economy

Improve agricultural output & income

Objectives of the Project

1. Develop AI-based recommendation system
2. Predict optimal N, P, K values
3. Use machine learning for better accuracy
4. Adjust recommendations using soil test
5. Convert nutrients to real fertilizers (Urea, DAP, MOP)
6. Support environmentally friendly farming

Dataset Overview

Dataset Source:

- Based on Recommended Dose of Fertilizer(RDF) values
- Derived from Agridataset Source
- Inspired by data from UAS Bangalore research

Dataset Statistics:

Approximately :150 crops total of 300 samples

- Input features:

2 (Crop, Condition: Irrigated/Rainfed)

- Output features:

3 (Nitrogen, Phosphorus, Potassium)

- No missing values

Data Types:

Categorical: Crop, Condition

Numerical: N, P, K values

Dataset Example

Crop	Condition	N (kg/ha)	P (kg/ha)	K (kg/ha)
Rice	Irrigated	100	60	40
Wheat	Rainfed	80	50	30
Maize	Irrigated	120	70	50
Cotton	Rainfed	90	55	35

RDF: Recommended Dose of Fertilizers - Used to train ML model

The model learns patterns to provide customized recommendations for different crops and soil conditions

⚙ Methodology / Approach

1. **Data Collection**
   Collect crop and fertilizer data

2. **Data Preprocessing**
   Convert categorical data to numbers

3. **Model Training**
   Use Random Forest Regressor

4. **Prediction**
   Predict NPK from inputs

5. **Soil Adjustment**
   Modify based on soil test

6. **Fertilizer Conversion**
   Convert NPK → Urea, DAP, MOP

Data Cleaning & Preprocessing:

Dataset checked for consistency

No missing values present

Categorical feature s (crop, condition) converted using Label Encoding

Data structured into input (X) and output (Y)

Exploratory Data Analysis (EDA):

Observed variation of N, P, K across different crops

Identified patterns in fertilizer requirements

Understood relationship between crop type and nutrient needs

Model / Algorithm Selection:

Selected Random Forest Regressor

Suitable for predicting continuous values (NPK)

Handles non-linear relationships

Reduces overfitting compared to single Decision Tree

Model Training and Validation:

Model trained using crop and condition as inputs

Output variables: N, P, K values

Dataset split conceptually into training and testing (if asked)

Model learns patterns from RDF dataset

Evaluation Metrics

Since this is a regression problem, accuracy is not used

Metrics used:MAE (Mean Absolute Error)

RMSE (Root Mean Squared Error),R^2 Score

These measure prediction error and model performance

🧰 Tools & Technologies Used

Python

Programming Language

Pandas

Data Handling & Analysis

Scikit-learn

Machine Learning Library

Google Colab

Development Platform

Random Forest

ML Algorithm

Results

Model Performance:

✓ Successfully predicts NPK values

✓ Provides customized recommendations

✓ Reduces excess fertilizer usage

Key Benefits:

- 💰 Saves cost for farmers
- 🌾 Improves crop yield
- ♻️ Promotes sustainability
- 📊 Data-driven approach

Output: NPK values (kg/ha) + Fertilizer quantities (Urea, DAP, MOP)

MODEL PERFORMANCE (RDF)

N

R2: 0.581

MAE: 12.29

RMSE: 23.53

P

R2: 0.63

MAE: 4.592

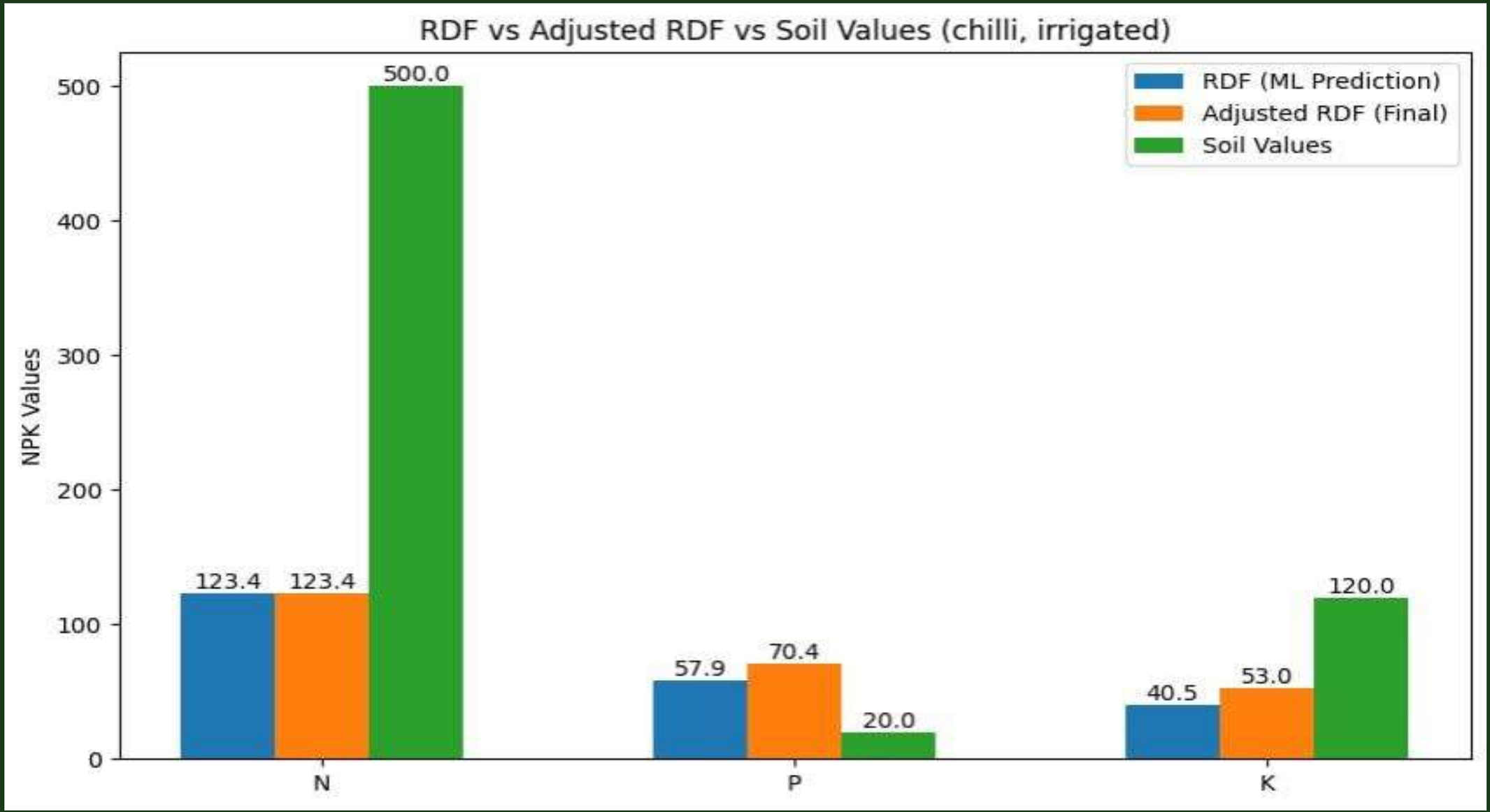
RMSE: 6.238

K

R2: 0.565

MAE: 4.381

RMSE: 7.341


RDF vs Adjusted RDF vs Soil Values (chilli, irrigated)
RDF (ML Prediction)
Adjusted RDF (Final)
Soil Values
NPK Values
0
100
200
300
400
500
123.4
123.4
500.0
57.9
70.4
20.0
40.5
53.0
120.0
N
P
K

Nutrient	RDF (kg ha^{-1})	Addition for low fertility rating (kg ha^{-1})	Deductions for high fertility rating (kg ha^{-1})
Nitrogen	< 50	No changes	No changes
	51 - 100	+ 12.5	- 12.5
	101 - 175	+ 25.0	- 25.0
	176 - 250	+ 37.5	- 37.5
	251 - 325	+ 50.0	- 50.0
Phosphorus	< 25	No changes	No changes
	26 - 75	+ 12.5	- 12.5
	76 - 125	+ 25.0	- 25.0
Potassium	< 25	No changes	No changes
	26 - 50	+ 12.5	- 12.5
	51 - 100	+ 25.0	- 25.0
	101 - 175	+ 37.5	- 37.5

Nutrient	Low	Medium	High
Nitrogen (N) (kg/ha)	< 280	280 – 560	> 560
Phosphorus (P_2O_5) (kg/ha)	< 22.9	22.9 – 56.33	> 56.33
Potassium (K_2O) (kg/ha)	141	141 – 336	> 336

Discussion / Analysis

Insights	Challenges	Limitations
• AI improves accuracy	• Small dataset size	• Works for trained crops only
• Better nutrient management	• No weather data	• Accuracy depends on data quality
• Data-driven farming	• Requires soil testing	• Need real-time updates

✓ Conclusion & Future Work

Conclusion

- ✓ AI transforms agriculture
- ✓ Precise recommendations
- ✓ Improves productivity & health

Future Work

- → Mobile app for farmers
- → IoT sensor integration
- → Weather predictions
- → Larger datasets
- → Deep Learning models

References & Acknowledgements

- Faculty guidance and support
- Open-source datasets
- Libraries: Scikit-learn, Pandas, NumPy
- Google Colab platform
- Team members contribution
- UAS Bengaluru resources

Thank you!